

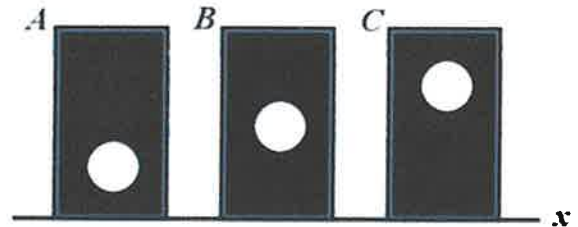
2.4 A general shape with an area of 20 in^2 has a moment of inertia of 106.7 in^4 with respect to the x axis. The centroid is located 2 inches above the x axis. What is the moment of inertia with respect to a centroidal x' axis?

$$I_x = \bar{I} + A d_y^2$$

$$\bar{I} = I_x - A y^2 = 106.7 - (20)(2)^2$$

$$\boxed{\bar{I} = 26.7 \text{ in}^4}$$

2.5 The figure shows three areas. Each area is a rectangle with a missing circle. The three rectangles have the same dimensions, and the three missing circles have the same dimensions. Which area has the largest moment of inertia about the x axis?



$$I_x = \int_A y^2 dA$$

I_x for A is largest.

Section 3 – Product of Inertia

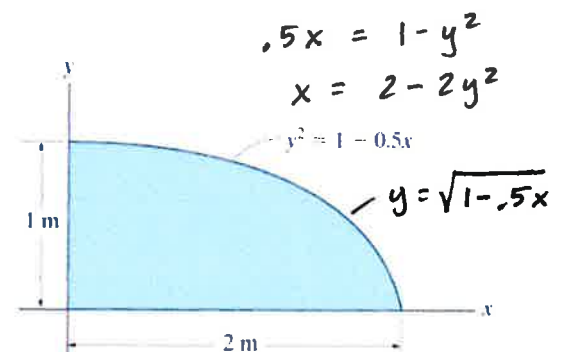
3.1 When using the parallel axis theorem to determine the product of inertia for an area about a set of non-centroidal x - y axes, $I_{xy} = \bar{I}_{x'y'} + A d_x d_y$, the second term on the right side of the equation is:

- A. Area times the differential widths dx and dy
- ☒ B. Area times the distances between the x - y origin and the x' - y' origin.
- C. Always positive
- D. Always negative
- E. Always zero

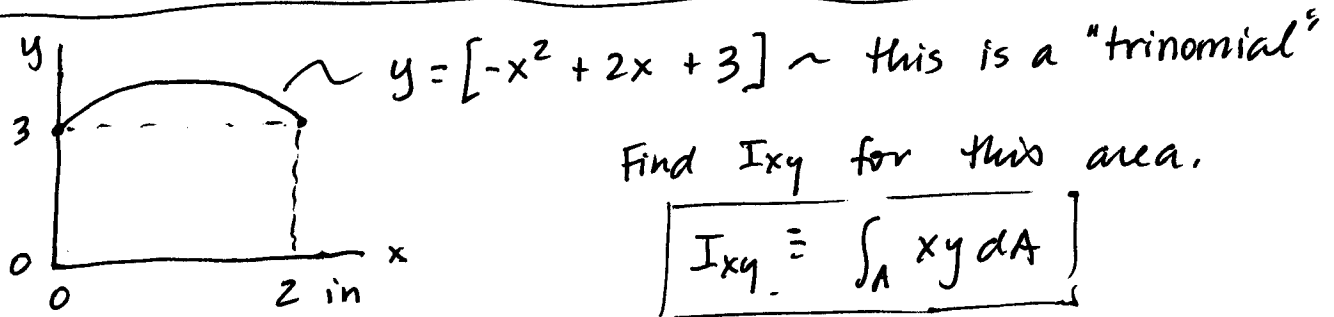
3.2 Determine the product of inertia of the shaded area with respect to the x and y axes.

$$I_{xy} = \int_0^2 \int_0^{\sqrt{1-.5x}} xy \, dy \, dx$$

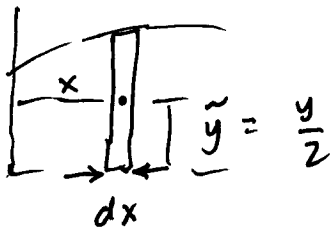
$$\begin{aligned} \text{TI89: } & \int \left(\int (x * y, y, 0, \sqrt{1-.5 * x}) \right), x, 0, 2) \\ & = \boxed{.333 \text{ m}^4 = I_{xy}} \end{aligned}$$



POI (I_{xy}) Integration: for a Trinomial using TI-89



Using a single integral:



$$dA = (-x^2 + 2x + 3) \, dx$$

$$x = x$$

$$y = y/2$$

$$I_{xy} = \int_0^2 x \left(\frac{-x^2 + 2x + 3}{2} \right) (-x^2 + 2x + 3) \, dx$$

To solve this by hand need to multiply (square)
the trinomial... Painful!

On the TI-89: $\int (.5 * x * (-x^2 + 2 * x + 3)^2, x, 0, 2)$

$$I_{xy} = 13.53 \, \text{in}^4$$

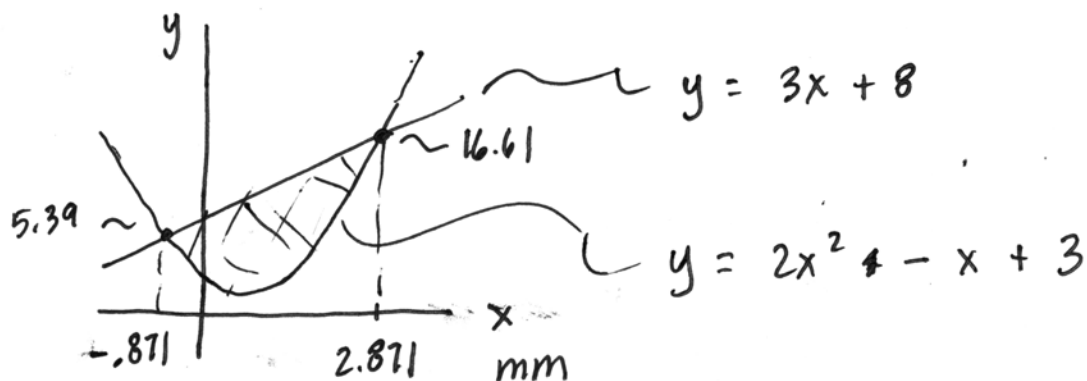
Using a double integral:

$$I_{xy} = \int_0^2 \int_0^{-x^2 + 2x + 3} xy \, dy \, dx$$

} End up having
to square it again!
Painful!

TI-89: $I_{xy} = \int (\int (x * y, y, 0, -x^2 + 2 * x + 3), x, 0, 2)$

$$I_{xy} = 13.53 \, \text{in}^4$$



$$I_{xy} = \int_A xy \, dA$$

$$I_{xy} = \int_{-0.871}^{2.871} \int_{2x^2 - x + 3}^{3x + 8} xy \, dy \, dx$$

TI-89

$$I_{xy} = \int \left(\int (x * y, y, 2 * x^2 - x + 3, 3 * x + 8), x, -0.871, 2.871 \right)$$

$$I_{xy} = 179.85 \, \text{mm}^4$$